Muhammad Taha Mukhtar

BS Computer Science | National University of Sciences and Technology (NUST)
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PROFESSIONAL PROFILE

A dedicated Computer Science student with a strong passion for math, problem-solving, and programming who has been involved with renowned international open-source communities and research groups, collaborating on innovative projects, and delivering impactful solutions for societal problems. Proficient in Full-stack development and deep learning, I use technology to address difficult problems in various domains of science.

EDUCATION

- BS in Computer Science School of Electrical Engineering and Computer Science , 2.99

WORK & INTERNSHIP EXPERIENCE

Google Summer of Code -- CVAT.ai

01-May-2024 - 27-Aug-2024

- Improved the CVAT platform by adding a user-configurable shortcut system, enhancing annotation efficiency.
- Tested new shortcut features with Cypress, achieving 100% test coverage.
- Contributed to integrating Meta’s SAM into CVAT.ai’s codebase beyond the program’s scope.

Deutsches Forschungszentrum für Künstliche Intelligenz

03-Jun-2024 - 01-Aug-2024

- Engineered a semi-supervised model for detecting informal settlements using remote sensing.
- Applied techniques like FixMatch, domain adaptation between cies and pseudo-label refinement via SAM2.
- Leveraged Germany's largest deep learning infrastructure with H100/A100 GPUs for advanced AI research.

PROJECTS

Slum-i, Research Project

- Worked on a €80,000 DAAD-funded project in collaboration with RPTU & DFKI, where I interned in Germany.
- Aligned with 3 UN SDGs, focusing on improving living conditions for slum dwellers in Pakistan.
- Upcoming publication in a top remote sensing journal, using SSL for segmenting informal settlements.

lang-source, Open Source Project

- Created a CLI utility using the Gemini API to automate localisation workflows, supporting 12 languages and reducing manual effort in creating JSON/YAML language files.
- Published as an npm package with 500+ downloads and has 5 Stars on GitHub.

TECHNICAL SKILLS

- Digital marketing
- Typescript
- Javascript (es6)
- Mongodb (database)
- Docker

- Mern stack
- Aws
- Tensorflow/pytorch

SOFT SKILLS

- Strategic communications